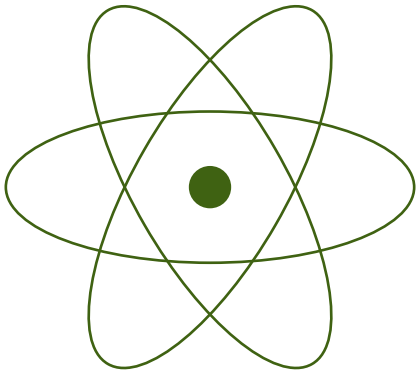
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Dynamics I

Fundamental Forces and Newton's Laws

Olympiad Training • Mechanics

SPhOH Team

FIRST EDITION • 2026

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About This Problem Set

This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused and substantial collection of core problems for deliberate problem-solving practice in classical dynamics.

The handout is **problem-centered**. Explanations are intentionally brief and are included only where they help establish the ideas or techniques needed for the problems.

How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

For difficult problems, several progressive hints may be provided. Try to use the least amount of assistance necessary before returning to the problem.

Difficulty Levels

The problems are divided into training levels:

- **Level I** — Direct application of Newton's laws and force balancing.
- **Level II** — Multi-body systems, friction, and inclined planes.
- **Level III** — Rigorous national/international Olympiad level dynamics.

Find more at: `https://devfhd.github.io/SPhOH/`.

CHAPTER 1

Dynamics I

Topic Note: Newton's Laws and Fundamental Forces
Dynamics links the motion of bodies to the forces causing it. Key principles include:

- **Newton's First Law (Law of Inertia):** A body remains at rest or continues to move in a straight line at constant velocity unless acted upon by a net external force:

$$\vec{F}_{\text{net}} = 0 \implies \vec{a} = 0 \quad (\vec{v} = \text{constant})$$

- **Newton's Second Law:** The net force acting on a body equals the rate of change of its momentum:

$$\vec{F}_{\text{net}} = \frac{\Delta \vec{p}}{\Delta t} = m\vec{a} \quad \text{(for constant mass)}$$

- **Newton's Third Law (Action-Reaction):** When one body exerts a force on a second body, the second body simultaneously exerts a force equal in magnitude and opposite in direction on the first body:

$$\vec{F}_{A\to B} = -\vec{F}_{B\to A}$$

- **Friction Forces:** Static friction opposes impending motion ($f_s \leq \mu_s N$), while kinetic friction opposes active sliding ($f_k = \mu_k N$).
- **Tension and Constraints:** Ideal strings are massless and inextensible, transmitting uniform tension forces along their length and coupling accelerations of connected bodies.

1.1 Static Equilibrium and Cable Tension

PROBLEM 1 **LEVEL I**

Vertical Cable Tension

A heavy light-fixture of mass $m = 12.0$ kg hangs at rest from a single vertical cord connected to the ceiling.

Find the tension T in the cord.

PROBLEM 2 **LEVEL I**

Horizontal Push on a Wall

A box of mass $m = 5.0$ kg is held at rest on a smooth horizontal floor. A person pushes horizontally on the box with a force $F = 20$ N towards a rigid vertical wall.

Determine the magnitude of the normal force exerted by the wall on the box.

PROBLEM 3 LEVEL I

Single Cable Weight Balance

A crate of mass $m = 25.0$ kg hangs vertically at rest from a single vertical steel cable attached to an overhead crane hook.

Determine the tension force T in the cable.

PROBLEM 4 LEVEL I

Crate of Toys on an Accelerating Sleigh

A crate of toys remains at rest on a sleigh as the sleigh is pulled up a hill with an increasing speed. The crate is not fastened down to the sleigh.

Which force is directly responsible for the crate's increase in speed up the hill?

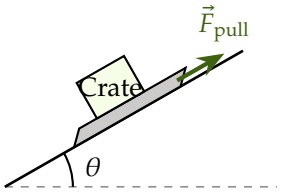
(a) The force of static friction of the sleigh on the crate

(b) The contact force (normal force) of the ground on the sleigh

(c) The contact force (normal force) of the sleigh on the crate

(d) The gravitational force acting on the sleigh

(e) No force is needed

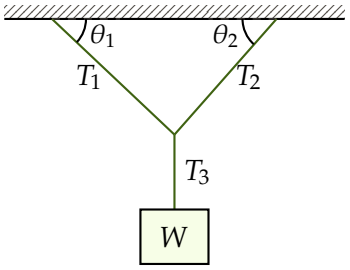


PROBLEM 5 LEVEL I

Suspended Cargo in Equilibrium

A crate of industrial supplies weighing $W = 360$ N is held in static equilibrium by three light cables joined at a central node. Two upper cables extend to a rigid overhead ceiling, forming angles $\theta_1 = 53.0°$ and $\theta_2 = 37.0°$ relative to the horizontal.

Determine the tension forces T_1, T_2, and T_3 in each supporting cable.



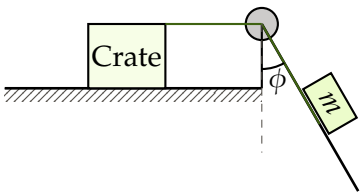
PROBLEM 6

LEVEL I

Stationary Container on Incline with Pulley

A heavy crate rests in static equilibrium on a horizontal table. A light, flexible cord attached to the crate extends horizontally over a smooth, frictionless pulley and connects to a secondary block of mass $m = 3.00$ kg that rests on an attached smooth ramp inclined at an angle $\phi = 30.0°$ relative to the vertical.

Calculate the tension force T exerted by the cord on the stationary crate.



PROBLEM 7

LEVEL II

General Cable Tension Formula

A payload of weight F_g hangs in static equilibrium suspended by three ideal wires. The two upper support wires attach to a horizontal ceiling at inclination angles θ_1 and θ_2 measured relative to the horizontal.

Prove analytically that the tension T_1 in the left wire is given by:

$$T_1 = \frac{F_g \cos\theta_2}{\sin(\theta_1 + \theta_2)}$$

PROBLEM 8 LEVEL II

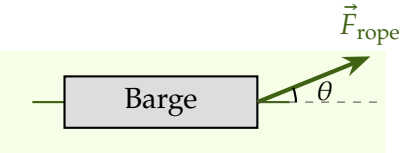
Towing a River Barge

A draft horse hauls a canal barge of mass $M = 8800$ kg along a straight waterway parallel to the positive x-axis. The towline exerts a tension force of magnitude $F = 6500$ N oriented at an angle $\theta = 22.0°$ relative to the positive x-axis. The barge accelerates forward along the x-axis at $a = 0.150$ m/s^2.

Determine:

(a) The magnitude of the net water resistance force acting on the barge.

(b) The direction angle of this water force relative to the positive x-axis.



PROBLEM 9 LEVEL II

Terminal Velocity Under Quadratic Drag

A spherical raindrop of mass $m = 0.15$ g and cross-sectional area $A = 2.0 \times 10^{-4}$ m^2 falls vertically through the atmosphere. Air resistance exerts a drag force governed by the quadratic drag formula:

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

where the air density is $\rho = 1.20$ kg/m^3 and the drag coefficient is $C_d = 0.47$.

Determine the terminal velocity v_t reached by the falling raindrop.

PROBLEM 10 LEVEL III

Charged Particle in Uniform Electric Field on an Incline

A small block of mass $m = 0.20$ kg carrying a positive charge $q = 15$ μC rests on a smooth, non-conducting plane inclined at an angle $\theta = 30.0°$ to the horizontal. A uniform electric field $\vec{E}$ of magnitude $E = 4.0 \times 10^4$ N/C is directed horizontally to the right (parallel to the base of the incline).

The electrostatic force on a charge in an electric field is given by $\vec{F}_e = q\vec{E}$.

Determine:

(a) The acceleration of the charged block up or down the incline.

(b) The magnitude of the normal force exerted by the incline on the block.

PROBLEM 11 LEVEL III

Dockside Crane Rigging Tension Analysis

A crane lifts a container of mass $M = 2.80 \times 10^4$ kg. Four identical lower cables, each with tension T_4, support the container and make an angle $\theta_4 = 60.0°$ with the vertical. They connect to a horizontal spreader bar supported by two identical cables of tension T_2, each making an angle $\theta_2 = 40.0°$ with the vertical. A vertical main cable has tension T_1. Neglect the mass of the bar. Find (a) T_1, (b) T_2, and (c) T_4.

1.2 Multi-Body Dynamics and Coupled Systems

PROBLEM 12 LEVEL I

Single Accelerated Block

A crate of mass $m = 8.0$ kg rests on a frictionless horizontal floor. A constant horizontal force $F = 24.0$ N pulls the crate to the right.

Calculate the magnitude of the crate's acceleration.

PROBLEM 13 LEVEL I

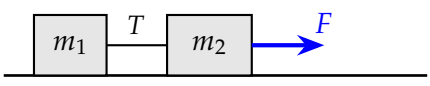
Two Blocks Connected by a String

Two blocks of masses $m_1 = 2$ kg and $m_2 = 3$ kg rest on a frictionless horizontal surface. A horizontal force $F = 15$ N is applied to block m_2, which is connected to m_1 by a light string.

Find:

(a) The acceleration of the system.

(b) The tension T in the string connecting the two blocks.



PROBLEM 14 LEVEL I

Simple Ideal Pulley System

A block of mass $m_1 = 4.0$ kg rests on a smooth horizontal table and is attached to a light cord that passes over a frictionless pulley. The other end of the cord supports a vertically hanging mass $m_2 = 2.0$ kg.

Determine the magnitude of the acceleration of the system when released from rest.

PROBLEM 15 LEVEL I

Two Blocks Pushed on a Frictionless Surface

Two blocks with masses $m_1 = 3.0$ kg and $m_2 = 2.0$ kg are placed in contact on a horizontal frictionless surface. A horizontal force $F = 15.0$ N pushes block m_1 directly.

Find:

(a) The acceleration of the two blocks.

(b) The contact force between block m_1 and block m_2.

PROBLEM 16 LEVEL I

Connected Blocks on a Frictionless Surface

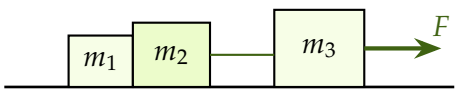
Three blocks with masses $m_1 = 1.5$ kg, $m_2 = 2.5$ kg, and $m_3 = 4.0$ kg are placed on a smooth, horizontal track. Blocks m_1 and m_2 are in contact, and m_2 is connected to m_3 by a light cord. A constant horizontal force $F = 48.0$ N pulls the lead block m_3 forward.

Calculate:

(a) The common acceleration of the combined three-block assembly.

(b) The tension in the coupling cord between m_3 and m_2.

(c) The contact force transmitted between m_2 and m_1.



PROBLEM 17 LEVEL II

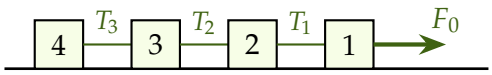
Tension in a Train of 4 Connected Blocks

A collection of 4 identical blocks, each of mass m, are connected by unstretchable ropes of negligible mass on a horizontal surface. An external force F_0 acts on Block 1, pulling horizontally to the right.

Find the tension T_1, T_2, and T_3 in the ropes connecting the blocks for:

(a) The case where there is no friction between the blocks and the surface.

(b) The case where the coefficient of kinetic friction between each block and the surface is μ_k (assuming F_0 is large enough to accelerate the blocks).



PROBLEM 18 LEVEL II

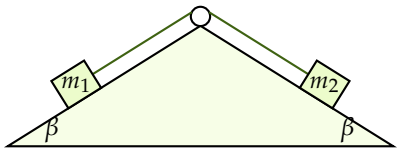
Symmetrical Double-Incline System

Two blocks of masses $m_1 = 4.00$ kg and $m_2 = 7.50$ kg are connected by a light string passing over a frictionless apex pulley atop a double inclined wedge. Both inclined faces form an angle $\beta = 32.0°$ with the horizontal and are frictionless.

Determine:

(a) The magnitude and direction of the acceleration of each block.

(b) The tension force in the connecting string.



PROBLEM 19 LEVEL II

Electrostatic Repulsion of Suspended Charges

Two identical small spheres, each of mass $m = 5.0$ g, are suspended from a common point by two light insulating strings of length $L = 0.50$ m. When given identical positive charges q, the spheres repel each other and hang in equilibrium separated by a horizontal distance $d = 6.0$ cm.

Using Coulomb's law for electrostatic force, $F_e = k_e \frac{q^2}{r^2}$, determine the magnitude of the charge q on each sphere. (Take $k_e = 8.99 \times 10^9$ N m^2/C^2 and $g = 9.81$ m/s^2).

PROBLEM 20 LEVEL II

Identical Mass Atwood System with Table Friction

Three identical blocks, each of mass m, are connected as shown: block m_0 hangs vertically from an ideal string passing over a frictionless pulley at the edge of a horizontal surface, while blocks m_1 and m_2 lie in tandem on the horizontal surface, linked by a second string. The coefficient of friction between the horizontal surface and blocks m_1 and m_2 is k.

(a) Determine the downward acceleration a of mass m_0.

(b) Find the tension T_{12} in the connecting cord between masses m_1 and m_2.

(c) Specify the threshold condition on k for which motion occurs.

PROBLEM 21 LEVEL II

Contact Force Between Adjacent Blocks on an Incline

Two rectangular blocks 1 and 2 with masses m_1 and m_2 are placed in direct contact on an inclined plane forming an angle α with the horizontal. The coefficients of friction between the incline and blocks 1 and 2 are k_1 and k_2, respectively.

(a) Find the condition under which the blocks push against each other during motion down the incline.

(b) Determine the acceleration of the combined blocks and the normal force of contact between them when this condition is satisfied.

PROBLEM 22 LEVEL II

Atwood Machine in an Accelerating Elevator

A frictionless, light pulley is attached to the ceiling of an elevator cabin. An ideal string passing over the pulley supports two masses, m_1 and m_2. The elevator starts ascending with a constant upward acceleration a_0.

(a) Calculate the acceleration of mass m_1 relative to the elevator car and relative to the fixed shaft frame.

(b) Determine the total vertical force exerted by the pulley support on the elevator ceiling.

PROBLEM 23 LEVEL III

Accelerated Double-Pulley Atwood System

A light, frictionless pulley A is suspended from a rigid ceiling. A light, inextensible string passes over pulley A, carrying a block of mass m_1 at one end and a second light, frictionless pulley B at the other end.

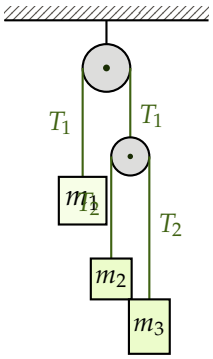
A second light string passes over pulley B and supports two blocks with masses m_2 and m_3 ($m_3 > m_2$). The entire system is released from rest in a uniform gravitational field g.

(a) Determine the kinematic constraint equations relating the accelerations a_1, a_2, and a_3 of the three masses with respect to the ground frame.

(b) Derive the expression for the acceleration a_1 of mass m_1 in terms of m_1, m_2, m_3, and g.

(c) Find the tension forces T_1 and T_2 in both strings.

(d) State the condition on the masses m_1, m_2, and m_3 for which pulley B remains stationary in space upon release.



PROBLEM 24 LEVEL III

Table-Edge Pulley System with Nested Second Pulley

Mass m_1 hangs from a string over a light fixed pulley P_1. The string supports a second light pulley P_2. A second string passes around P_2, with one end attached to a wall and the other to mass m_2 on a smooth horizontal table.

(a) Draw a figure.

(b) Relate accelerations a_1 and a_2.

(c) Find the string tensions in terms of m_1, m_2, g.

(d) Find both accelerations in terms of m_1, m_2, g.

PROBLEM 25 LEVEL III

Accelerated Block Assembly on a Moving Block

A large block of mass M rests on a frictionless horizontal plane. Mounted on top of M is a frictionless horizontal ledge terminating in a pulley, which suspends a hanging mass m_1 vertically along the right side of M. A second mass m_2 rests on top of M and connects to m_1 via a light cord running over the pulley.

Determine the horizontal driving force F that must be applied to M so that masses m_1 and m_2 remain completely stationary relative to M.

PROBLEM 26 LEVEL III

Modified Double-Pulley System on an Incline

A block of mass m_1 slides on a smooth plane inclined at angle α. A cord attached to m_1 runs parallel to the incline, passes over a fixed pulley at the top, and connects to a second block of mass $m_2 = \eta m_1$ hanging vertically.

Determine the acceleration of block m_2 for all possible cases depending on the mass ratio η and inclination α.

PROBLEM 27 LEVEL III

Triple Mass System with Compound Pulley

In a frictionless mechanism, mass m_0 rests on a smooth horizontal table, connected via strings and pulleys to two suspended masses m_1 and m_2.

Find the acceleration of mass m_1 in terms of m_0, m_1, m_2, and g.

1.3 Friction, Sliding, and Inclines

PROBLEM 28 LEVEL I

Normal Force on an Incline

A block of mass $m = 6.0$ kg rests on a smooth incline angled at $\theta = 30.0°$ relative to the horizontal.

Calculate the magnitude of the normal force N exerted by the incline on the block.

PROBLEM 29 LEVEL I

Kinetic Friction on a Flat Surface

A box of mass $m = 10.0$ kg slides across a horizontal rough floor with an initial speed $v_0 = 5.0$ m/s. The coefficient of kinetic friction between the box and the floor is $\mu_k = 0.25$.

Find the magnitude of the deceleration caused by friction.

PROBLEM 30 LEVEL I

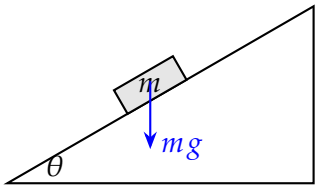
Block on an Inclined Plane

A block of mass $m = 4$ kg rests on a frictionless plane inclined at an angle $\theta = 30°$ to the horizontal.

Find:

(a) The component of the gravitational force acting down the incline.

(b) The acceleration of the block as it slides down the plane.



PROBLEM 31 LEVEL II

Threshold Friction under Normal Loading

A block of mass $m = 3.00$ kg rests on a horizontal table. A constant horizontal force $F = 8.00$ N is applied to it. In addition, a vertical downward force P is exerted on top of the block. The coefficients of friction between block and table are $\mu_s = 0.45$ and $\mu_k = 0.30$.

Determine the magnitude of the resulting friction force acting on the block when P is set to:

(a) $P = 5.00$ N

(b) $P = 12.0$ N

(c) $P = 20.0$ N

PROBLEM 32 LEVEL II

Cargo Cart and Incline Launch

A freight cart with an open top and mass $M = 60.0$ kg rolls freely on a level frictionless track to the left at speed $v_0 = 4.50$ m/s. A parcel of mass $m = 12.0$ kg slides down a chute angled at $\theta = 30.0°$ to the horizontal, exiting the chute's lower edge with a speed $u_0 = 4.00$ m/s. The exit point of the chute is located at height $h = 3.50$ m above the bed of the cart.

(a) Determine the speed of the parcel immediately before landing in the cart.

(b) Compute the final horizontal speed of the combined cart and parcel after collision.

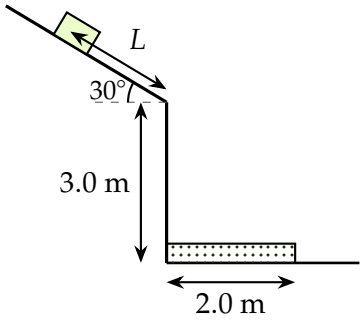
PROBLEM 33 LEVEL II

Box Sliding off a Roof into a Flower Bed

A box of nails begins sliding from rest on the roof of a house. The roof makes an angle of 30° above the horizontal. The coefficient of sliding friction is $\mu_k = 1/4$. The box slides off the edge of the roof with a speed of 3.5 m/s.

(a) What is the distance L the box slides on the roof before falling off? (Use $g = 9.8$ m/s^2).

(b) The flower bed extends from the side of the house and is 2.0 m wide. The edge of the roof is 3.0 m above the ground. Does the box of nails land in the flowers? Justify your answer with a calculation that shows where the box lands.



PROBLEM 34 LEVEL II

Non-Hookian Spring on an Incline

A non-Hookian spring exerts a restoring force $F = -kx^2$, where k is a positive constant and x is the displacement from its unstretched position. A mass m connected to this spring (initially unstretched and at rest) is released on a frictionless ramp inclined at 30° relative to the horizontal, with the cord passing over an ideal pulley.

How far x does the spring extend before the system momentarily comes to rest?

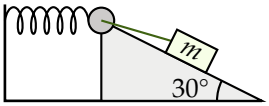
(a) $\left(\frac{3mg}{2k}\right)^{1/2}$

(b) $\left(\frac{mg}{k}\right)^{1/2}$

(c) $\left(\frac{2mg}{k}\right)^{1/2}$

(d) $\left(\frac{\sqrt{3}mg}{k}\right)^{1/3}$

(e) $\left(\frac{3\sqrt{3}mg}{2k}\right)^{1/3}$



PROBLEM 35 LEVEL II

Tabletop Rotating Puck with Hanging Cylinder

A puck of mass $m = 1.50$ kg moves in a circle of radius $r = 20.0$ cm on a frictionless table. A cord through a hole in the table connects it to a hanging cylinder of mass $M = 2.50$ kg. What puck speed keeps the cylinder at rest?

PROBLEM 36 LEVEL II

Rock Climber in a Chimney

A 49 kg climber is braced between two vertical rock walls. The static friction coefficients at her shoes and back are 1.2 and 0.80, respectively. She reduces her push until she is on the verge of slipping. (a) Draw her free-body diagram. (b) Find the horizontal push. (c) What fraction of her weight is supported by friction at her shoes?

PROBLEM 37 LEVEL III

Accident Investigation on an Inclined Roadway

A forensic accident investigator analyzes a rear-end collision on a downhill road sloping at an angle $\theta = 14.0°$ below the horizontal. Car A locked its brakes and skidded a distance $d = 28.0$ m down the slope before striking stationary Car B. Car A's initial velocity when braking began was $v_0 = 20.0$ m/s.

Calculate the impact velocity of Car A for the following road conditions:

(a) Dry asphalt with kinetic friction coefficient $\mu_k = 0.65$.

(b) Wet, leaf-covered asphalt with kinetic friction coefficient $\mu_k = 0.12$.

PROBLEM 38 LEVEL III

Crate Pressed Against a Truck

A crate of mass m rests against the flat vertical front face of a heavy delivery truck. The coefficient of static friction between the crate and the truck's front surface is μ.

Find the minimum horizontal forward acceleration a_{min} of the truck necessary to prevent the crate from slipping downward under gravity.

PROBLEM 39 LEVEL III

Stacked Block and Slab System

A heavy slab of mass $M = 12.0\,\text{kg}$ lies on a smooth horizontal floor. A block of mass $m = 8.00\,\text{kg}$ rests on top of the slab. A horizontal force $F = 110\,\text{N}$ is applied directly to the bottom slab. The coefficient of static friction between block and slab is $\mu_s = 0.40$, and kinetic friction is $\mu_k = 0.25$.

Analyze the motion: determine whether the block slides relative to the slab, and calculate the resulting accelerations a_{slab} and a_{block}.

PROBLEM 40 LEVEL III

Tabletop Incline Projectile Launch

A small block of mass $m = 1.50\,\text{kg}$ is released from rest at a vertical height $h = 0.600\,\text{m}$ above a tabletop, sliding down a smooth ramp inclined at $\theta = 30.0^\circ$. The bottom of the ramp sits at the edge of the tabletop, which is elevated at height $H = 1.80\,\text{m}$ above the floor.

(a) Determine the acceleration of the block along the ramp.

(b) Calculate the velocity vector of the block as it leaves the ramp edge.

(c) Find the horizontal range R from the table edge to where the block impacts the floor.

(d) Compute the total time elapsed from release until impact with the floor.

(e) State whether varying the mass m would alter any of these physical quantities.

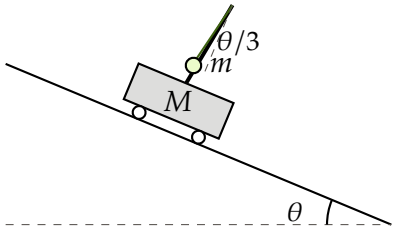
PROBLEM 41 LEVEL III

Accelerometer on an Inclined Cart

An accelerometer consists of a ball of mass m attached to one end of a string. The other end of the string is attached to a cart of mass M. When the cart rolls down an inclined plane of angle θ with respect to the horizontal, the string makes a constant angle of $\theta/3$ with respect to the normal to the inclined plane. Express your answers in terms of known and given quantities (g, θ). Assume air resistance is negligible.

(a) What is the acceleration of the cart down the incline?

(b) What is the coefficient of kinetic friction μ between the cart and the plane?



PROBLEM 42 LEVEL III

Friction-Coupled Rod and Sliding Ball

A smooth, light pulley supports a flexible cord. One end of the cord is attached to a vertical rod of mass M, and the other end carries a small heavy ball of mass m ($M > m$). The ball has a central bore allowing it to slide along the cord with a constant friction force f. Initially, the ball is positioned directly opposite the lower tip of the rod. Upon release from rest, both bodies begin moving with constant accelerations.

Find the magnitude of the friction force f if the ball reaches the upper tip of the rod of length L after a time interval Δt.

PROBLEM 43 LEVEL III

Time to Traverse Hanging Rod

A spherical mass m_1 is attached to a string passing over fixed pulleys and positioned level with the lower end of a vertical hollow pipe of mass $m_2 = \frac{1}{2}m_1$ and length $L = 1.20$ m. The system is released from rest.

Calculate the time required for mass m_1 to reach the level of the upper end of the pipe.

PROBLEM 44 LEVEL III

Maximum Ascent of Pulley System Mass

In a dual-pulley setup, mass m_1 is $n = 3.5$ times heavier than mass m_2. Mass m_2 is initially situated at height $h = 25.0$ cm above a stopping surface. The system is released from rest.

Compute the maximum vertical height above its initial position reached by body m_2.

CHAPTER 2

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

Hint 1 | Problem 1 – Vertical Cable Tension

Apply vertical equilibrium $\sum F_y = 0$: $T - mg = 0$.

Hint 2 | Problem 2 – Horizontal Push on a Wall

In horizontal equilibrium ($\sum F_x = 0$), the normal force from the wall balances the applied push force: $N = F$.

Hint 3 | Problem 3 – Single Cable Weight Balance

Apply vertical static equilibrium $\sum F_y = 0$, balancing tension T against the gravitational weight $W = mg$.

Hint 4 | Problem 4 – Crate of Toys on an Accelerating Sleigh

Consider the forces acting parallel to the incline specifically on the crate. The only contact force parallel to the sleigh's surface preventing relative sliding and providing upward acceleration is static friction.

Hint 5 | Problem 5 – Suspended Cargo in Equilibrium

Set up equilibrium conditions for the central node ($\sum F_x = 0$ and $\sum F_y = 0$). Solve the system of linear equations for T_1 and T_2.

Hint 6 | Problem 6 – Stationary Container on Incline with Pulley

Examine the equilibrium of the block on the smooth incline. The tension along the ideal string must balance the component of the block's weight parallel to the incline: $T = mg \sin\theta = mg \cos\phi$.

Hint 7 | Problem 7 – General Cable Tension Formula

Express T_2 in terms of T_1 using horizontal equilibrium ($T_1 \cos\theta_1 = T_2 \cos\theta_2$). Substitute into vertical equilibrium ($T_1 \sin\theta_1 + T_2 \sin\theta_2 = F_g$) and apply the trigonometric identity $\sin(\theta_1 + \theta_2) = \sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2$.

Hint 8 | Problem 8 – Towing a River Barge

Apply Newton's second law along x ($F\cos\theta + F_{\text{water},x} = Ma$) and y ($F\sin\theta + F_{\text{water},y} = 0$). Find the resultant magnitude and direction using vector components.

Hint 9 | Problem 9 – Terminal Velocity Under Quadratic Drag

At terminal velocity, acceleration is zero, so the gravitational force balances the drag force: $mg = \frac{1}{2}\rho v_t^2 C_d A$. Solve for v_t.

Hint 10 | Problem 10 – Charged Particle in Uniform Electric Field on an Incline

Resolve the horizontal electric force into components parallel and perpendicular to the incline. Parallel to the incline, the net force is $F_{\parallel} = qE\cos\theta - mg\sin\theta = ma$. Perpendicularly, $N = mg\cos\theta + qE\sin\theta$.

Hint 11 | Problem 11 – Dockside Crane Rigging Tension Analysis

Vertical equilibrium gives $4T_4\cos\theta_4 = Mg$ for the container, $2T_2\cos\theta_2 = 4T_4\cos\theta_4 = Mg$ for the bar, and $T_1 = 2T_2\cos\theta_2 = Mg$ for the main cable.

Hint 12 | Problem 12 – Single Accelerated Block

Apply Newton's second law along the horizontal axis: $F = ma$.

Hint 13 | Problem 13 – Two Blocks Connected by a String

Treat both blocks together to find the system acceleration first, then isolate one block to find the tension.

Hint 14 | Problem 14 – Simple Ideal Pulley System

Consider both blocks as a single system of mass $(m_1 + m_2)$ driven by the weight of the hanging block $m_2 g$.

Hint 15 | Problem 15 – Two Blocks Pushed on a Frictionless Surface

Calculate acceleration using total mass $a = \frac{F}{m_1+m_2}$. Then apply Newton's second law to m_2 alone to find contact force $N_c = m_2 a$.

Hint 16 | Problem 16 – Connected Blocks on a Frictionless Surface

Treat the three blocks as a single system of total mass $M_{\text{tot}} = m_1 + m_2 + m_3$ to find acceleration $a = F/M_{\text{tot}}$. Then isolate individual blocks to determine internal contact/tension forces.

Hint 17 | Problem 17 – Tension in a Train of 4 Connected Blocks

The total mass of the system is $4m$. Analyze the overall acceleration of the 4-block system first. Then, isolate trailing blocks to solve for each rope's tension (T_3 supports Block 4, T_2 supports Blocks 3 and 4, and T_1 supports Blocks 2, 3, and 4).

Hint 18 | Problem 18 – Symmetrical Double-Incline System

Write the equations of motion along each incline: $m_2 g \sin\beta - T = m_2 a$ and $T - m_1 g \sin\beta = m_1 a$. Sum them to isolate a.

Hint 19 | Problem 19 – Electrostatic Repulsion of Suspended Charges

Analyze the equilibrium of one sphere. The forces acting on it are tension T, gravity mg, and the electrostatic repulsion $F_e = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{d^2}$. Use geometry $\tan\theta = \frac{d/2}{\sqrt{L^2-(d/2)^2}}$ to equate force components: $F_e = mg\tan\theta$.

Hint 20 | Problem 20 – Identical Mass Atwood System with Table Friction

Write Newton's second law for the system of three equal masses m: $mg - 2kmg = 3ma$. For internal tension T_{12}, isolate block m_2: $T_{12} - kmg = ma$.

Hint 21 | Problem 21 – Contact Force Between Adjacent Blocks on an Incline

Block 2 tends to push block 1 if its unconstrained acceleration exceeds that of block 1, which requires $k_2 < k_1$. Set up coupled equations for m_1 and m_2 including the contact force N_c.

Hint 22 | Problem 22 – Atwood Machine in an Accelerating Elevator

Work in the non-inertial frame of the elevator, where an effective gravitational field $g_{\text{eff}} = g + a_0$ acts downward. The relative acceleration is $a_{\text{rel}} = \frac{m_1 - m_2}{m_1 + m_2} g_{\text{eff}}$.

Hint 23 | Problem 23 – Accelerated Double-Pulley Atwood System

Express the acceleration of m_2 and m_3 relative to pulley B as a_{rel}. Notice that pulley B is massless, so $T_1 = 2T_2$. Replace the system attached to pulley B with an effective reduced mass $m_{\text{eff}} = \frac{4m_2 m_3}{m_2 + m_3}$.

Hint 24 | Problem 24 – Table-Edge Pulley System with Nested Second Pulley

The length constraint gives $x_1 = 2x_2$, hence $a_1 = 2a_2$. Use $m_1 g - T_1 = m_1 a_1$, $T_2 = m_2 a_2$, and $T_1 = 2T_2$ for the movable pulley.

Hint 25 | Problem 25 – Accelerated Block Assembly on a Moving Block

For m_1 not to slide vertically relative to M, the string tension must balance gravity: $T = m_1 g$. For m_2 not to slide horizontally relative to M, it must accelerate horizontally with M at acceleration

$a = T/m_2 = \frac{m_1 g}{m_2}$. Apply $F = (M + m_1 + m_2)a$.

Hint 26 | Problem 26 – Modified Double-Pulley System on an Incline

Set up the net force along the direction of motion: $(m_2 - m_1 \sin\alpha)g = (m_1 + m_2)a$. Check the sign to determine if m_2 moves downward or upward.

Hint 27 | Problem 27 – Triple Mass System with Compound Pulley

Express the kinematic constraint relating the displacements of the three masses. Write Newton's second law for each body and solve the system of linear equations simultaneously.

Hint 28 | Problem 28 – Normal Force on an Incline

Resolve gravity into components. Perpendicular to the incline, equilibrium gives $N = mg\cos\theta$.

Hint 29 | Problem 29 – Kinetic Friction on a Flat Surface

The normal force is $N = mg$, making the kinetic friction $f_k = \mu_k mg$. Apply $f_k = ma$ to find acceleration.

Hint 30 | Problem 30 – Block on an Inclined Plane

Resolve the weight vector mg into components parallel and perpendicular to the inclined surface. Only the parallel component causes acceleration.

Hint 31 | Problem 31 – Threshold Friction under Normal Loading

Calculate the normal force $N = mg + P$ and the maximum static friction $f_{s,\max} = \mu_s N$. If $F \leq f_{s,\max}$, the block remains static and $f = F$. If $F > f_{s,\max}$, kinetic friction applies: $f = f_k = \mu_k N$.

Hint 32 | Problem 32 – Cargo Cart and Incline Launch

Use conservation of energy to find the speed of the parcel just before landing: $v_p = \sqrt{u_0^2 + 2gh}$. For the final cart speed, apply horizontal momentum conservation: $Mv_0 - mu_0\cos\theta = (M + m)V_f$.

Hint 33 | Problem 33 – Box Sliding off a Roof into a Flower Bed

For part (a), the acceleration along the roof is $a = g(\sin 30^\circ - \mu_k \cos 30^\circ)$. Use $v^2 = 2aL$. For part (b), calculate projectile flight time t using vertical displacement $y(t) = -3.0$ m with initial velocity components $v_{0x} = 3.5\cos 30^\circ$ and $v_{0y} = -3.5\sin 30^\circ$, then find horizontal distance $x = v_{0x}t$.

Hint 34 | Problem 34 – Non-Hookian Spring on an Incline

Apply the Work-Energy Theorem $\Delta K = W_{\text{net}} = 0$. The potential energy stored in a spring with force $F = -kx^2$ is $U_s = \int_0^x kx'^2 \mathrm{d}x' = \frac{1}{3}kx^3$. Work done by gravity down the incline is $W_g = mgx\sin 30^\circ = \frac{1}{2}mgx$. Equate $W_g = U_s$.

Hint 35 | Problem 35 – Tabletop Rotating Puck with Hanging Cylinder

For the stationary cylinder, $T = Mg$. The same tension supplies the puck's centripetal force, $T = mv^2/r$. Set the two expressions equal and solve for v.

Hint 36 | Problem 36 – Rock Climber in a Chimney

If the horizontal push is F_p, each normal force equals F_p. At impending slip, $f_{s,shoes} = 1.2F_p$ and $f_{s,back} = 0.80F_p$. Vertical equilibrium requires their sum to equal mg.

Hint 37 | Problem 37 – Accident Investigation on an Inclined Roadway

Acceleration down the incline is $a = g(\sin\theta - \mu_k\cos\theta)$. Use the kinematic relation $v^2 = v_0^2 + 2ad$ to calculate impact velocity v.

Hint 38 | Problem 38 – Crate Pressed Against a Truck

The normal force exerted on the crate by the accelerating truck is $N = ma$. For the crate not to slide downward, static friction must balance gravity: $f_s = mg \le \mu N = \mu ma$.

Hint 39 | Problem 39 – Stacked Block and Slab System

Find the maximum acceleration the block can sustain without slipping: $a_{\max} = \mu_s g$. Compare with the common acceleration of the combined system $a_{\text{comb}} = \frac{F}{M+m}$. If $a_{\text{comb}} > a_{\max}$, relative sliding occurs.

Hint 40 | Problem 40 – Tabletop Incline Projectile Launch

Ramp speed at exit is $v_0 = \sqrt{2gh}$. The block launches at angle $\theta = 30.0^\circ$ below the horizontal. Treat projectile motion using vertical displacement $-H = (-v_0\sin\theta)t - \frac{1}{2}gt^2$.

Hint 41 | Problem 41 – Accelerometer on an Inclined Cart

In the non-inertial frame of the accelerating cart, pseudo force $-m\vec{a}$ acts on the pendulum ball. Analyze components parallel and perpendicular to the incline to link string deflection angle $\theta/3$ to acceleration $a = g(\sin\theta - \mu\cos\theta)$.

Hint 42 | Problem 42 – Friction-Coupled Rod and Sliding Ball

Write Newton's second law for both bodies: $Mg - f = Ma_{\text{rod}}$ and $f - mg = ma_{\text{ball}}$. The relative acceleration is $a_{\text{rel}} = a_{\text{rod}} + a_{\text{ball}}$. Set $\frac{1}{2}a_{\text{rel}}(\Delta t)^2 = L$ to solve for f.

Hint 43 | Problem 43 – Time to Traverse Hanging Rod

Determine individual upward and downward accelerations from the pulley configuration constraint and Newton's second law. Find relative acceleration and set $s_{\text{rel}} = \frac{1}{2} a_{\text{rel}} t^2 = L$.

Hint 44 | Problem 44 – Maximum Ascent of Pulley System Mass

The motion consists of two phases: an accelerated phase while m_1 descends until it hits the floor, followed by a free-fall (upward projectile) phase for m_2 under gravity alone.